

AI Smart Traffic Intelligence Platform

Turning any camera feed into fair, evidence-based enforcement

Sri Lankan Students Innovation Challenge 2026 | Final

The problem

Sri Lanka records roughly **3,000 road deaths a year**, most from preventable violations.

Officers cannot watch every junction, 24 hours a day.

Violations recorded without photographic proof or a verified plate are disputed and unpaid.

Coverage and proof are the enforcement gap.

What we built

One AI pipeline on an ordinary camera detects **nine violation types**.

Reads number plates on every vehicle, not only offenders.

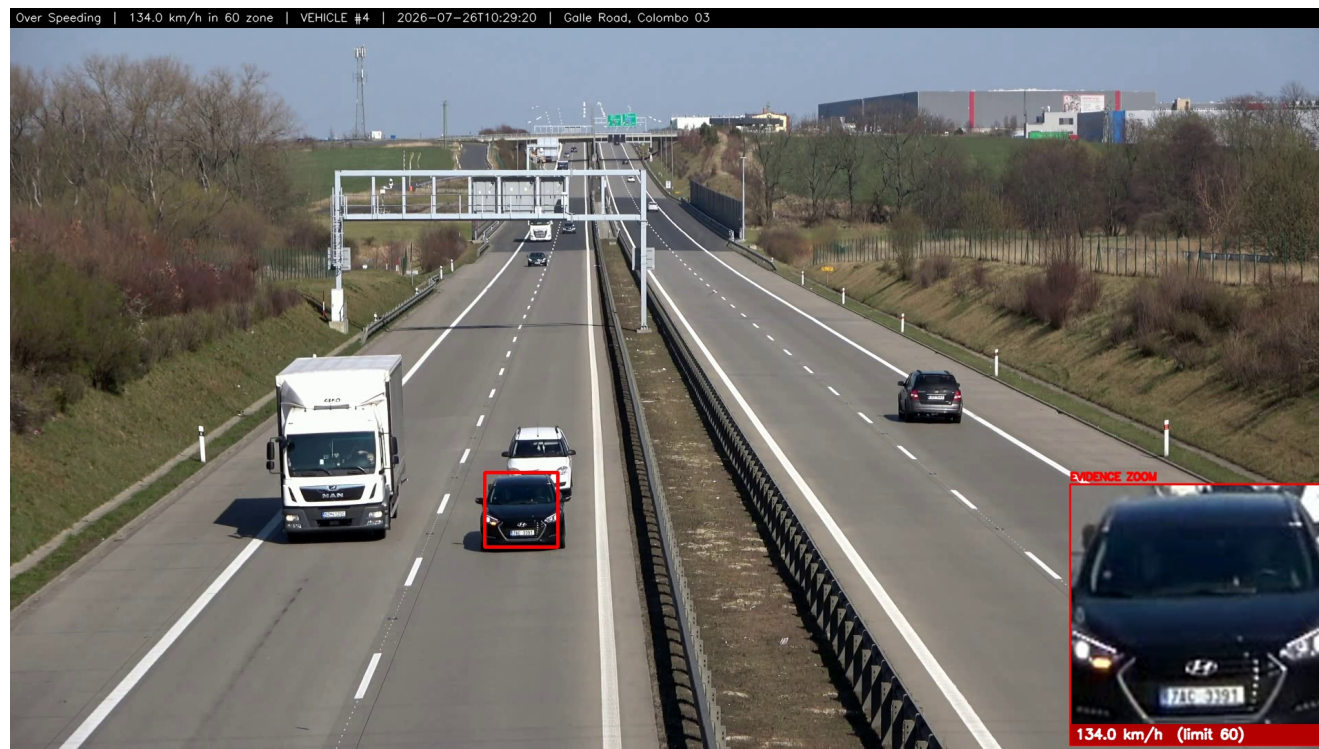
Measures **real speed** through camera calibration.

Generates an **e-challan** with stamped photographic evidence, delivered to the police.

Proof: over-speeding

Calibrated speed validated at **83-163 km/h** on highway footage.

Seven over-speeding events, each with court-ready evidence.



134.0 km/h in a 60 zone - vehicle boxed, plate zoomed, proof stamped on the image itself.

Proof: helmet and triple riding

Both violations detected from a **single uploaded photograph**, producing two separate challans.



Three riders and no helmets, judged on a CC-licensed public image.

Proof: number-plate recognition

Plates read on live Sri Lankan road footage.

A plate is confirmed only when **three separate reads agree**.



18 vehicles tracked; plate AAG 4002 read and confirmed; speed shown beside every vehicle ID.

Why it can be trusted

Parked vehicles are never fined - 41 vehicles, 0 violations on a parked-motorcycle clip.

Unreadable plates say UNREADABLE - a number is never invented.

No calibration means no speed shown, rather than a guess.

79 automated tests, including every case where a rule must *refuse* to fire.

We fixed six real false-positive defects found by testing on real footage.

Deployment

Works with **existing CCTV**, a phone stream, or a drone via OBS.

A pilot site runs on a single laptop; cities use GPU edge boxes.

Models for helmet, seatbelt, plate and three-wheeler are **drop-in files** - no code change.

Python, FastAPI, YOLOv8, ByteTrack, OpenCV, EasyOCR, SQLite.

Impact and roadmap

24/7 coverage without multiplying officers.

Court-ready evidence means fines that hold up when challenged.

Searchable plate log enables **repeat-offender** analytics.

Next: night and weather models, a vision-language model for ambiguous cases, and a live junction pilot.

Safer roads, proven by evidence.

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